

A Project Report on
Real-Time Sign Language Interpreter

Submitted to
**Dr.Manmohan Singh Bengaluru City University,
Bengaluru**



In Partial Fulfilment of the Requirement for the Award of Degree

Bachelor of computer application
in

Computer Science

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C E R T I F I C A T E

This is to certify that project work entitled “**Brain tumour detection using CNN and image processing**” has been successfully carried out by **Ms. Rakshitha M C, Mr. Joel R, Mr. Rajesh R S bearing** Reg Numbers U18UZ23S0080, U18UZ22S0081, U18UZ23S0079 students of Sixth Semester BCA. This Project is submitted in partial fulfilment for the award of degree in **Bachelor of Computer Applications** by **Dr Manmohan Bengaluru City University Bengaluru** for the academic year **2025-26**. It is certified that all corrections/suggestions indicated for internal assessment have been incorporated in the report deposited in the department. The project report has been approved as it satisfies the academic requirements in respect of project work prescribed for the said degree.

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DECLARATION

We, **Rakshitha M C, Joel R, Rajesh R S**, hereby declare that the project entitled **“Real-Time Sign Language Interpreter”** submitted to Dr. Manmohan Singh Bengaluru City University, Bengaluru in partial fulfilment for the Degree of **Bachelor of Computer Application in Computer Science** is the original work carried out by me under the guidance of **Sridhara J B, Department of Computer Science, Dr ManMohan Singh University Bengaluru City University**. This project or part there of has not been previously formed the basis for the award of other degree, diploma or other title in this or any other university.

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ABSTRACT

The Real-Time Sign Language Interpreter is a desktop application designed to bridge communication between the Deaf and Hearing communities. The system employs computer vision, deep learning, and natural language processing to recognise American Sign Language (ASL) gestures captured from a standard webcam and convert them into grammatically correct English sentences in real time.

The application operates in two complementary modes. In Letter Mode, a Dense Multi-Layer Perceptron (MLP) trained on 63-dimensional MediaPipe hand-landmark vectors classifies individual ASL fingerspelling letters with approximately 97% test accuracy. In Word Mode, a Bidirectional LSTM network augmented with a custom scaled dot-product Attention layer processes 60-frame temporal sequences of 126-dimensional two-hand landmark features, achieving 90–96% accuracy on trained vocabulary.

Recognised ASL keyword sequences are translated into fluent English using a dual-engine grammar system: a locally hosted Ollama Large Language Model (Mistral:instruct / Qwen3:4b) serves as the primary engine, with an EnhancedGrammar rule-based fallback handling pronoun normalisation, tense detection, verb conjugation, and compound-sign merging. Text-to-speech output is provided via a background-threaded TTSEngine (pyttsx3 or espeak), ensuring the main OpenCV user-interface loop is never blocked.

A complete data-collection and training pipeline is included, supporting up to 8× landmark augmentation for letters and 6× temporal augmentation for word sequences, enabling vocabulary expansion without external datasets. Performance benchmarking on a CPU-only Intel Core i7 laptop demonstrates end-to-end frame latency of 30–60 ms (16–33 fps), making the system practical for real-world use. The system operates entirely on-device, requiring no internet connectivity.

Keywords: American Sign Language, Real-Time Gesture Recognition, MediaPipe, Bidirectional LSTM, Attention Mechanism, Natural Language Processing, TensorFlow/Keras, Text-to-Speech, Accessibility.

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